Contribution Journal - Module 08 Lab: The Bias-Variance Tradeoff

In this module's lab, I explored model complexity, underfitting (high bias), and overfitting (high variance) using Python and scikit-learn in Google Colab. First, I generated a synthetic dataset based on a sine wave with added Gaussian noise and plotted the baseline distribution. Next, I built machine learning pipelines combining PolynomialFeatures and LinearRegression to train and compare three models of varying complexity: Degree 1, Degree 4, and Degree 15. By plotting their predictions against the original data, I verified that Degree 1 underfit the data, Degree 15 overfit the noise, and Degree 4 captured the true underlying sine wave pattern. I then implemented a custom learning curve plotting function using scikit-learn's learning_curve with 5-fold cross-validation (cv=5) to evaluate the training versus cross-validation scores across sample sizes for both the Degree 1 and Degree 15 models. Finally, I resolved single-line formatting syntax errors in the starter notebook, completed the knowledge check questions diagnosing the learning curves, and pushed the completed notebook to my GitHub repository.

AI Disclosure:
What AI was used: Google Gemini.
Why it was used: To help troubleshoot syntax errors caused by single-line code formatting in the starter notebook cells and to structure the pipeline code for polynomial regression and learning curves.
What I learned in reflection: I learned how polynomial degrees control model complexity, and how to read learning curves to diagnose whether a model is suffering from high bias (underfitting, where training and validation scores plateau low together) or high variance (overfitting, where a wide gap remains between high training and low validation scores).
How I modified it to make it my own: I formatted, executed, and verified all code blocks in my Google Colab environment, analyzed the resulting polynomial fit plots and 5-fold cross-validation learning curves, wrote out the knowledge check analysis, and committed the completed .ipynb file to my GitHub repository.